Chapter 3 · Energy and Speed Are Related

GUIDED NOTES

Name: _____ **Date:** _____

BQ How are energy and speed related?

My first thinking

V1 Vocabulary — Speed

Reader p. 13

speed — a measurement of the _____ an object travels over an amount of time.

N1 One Ball, Two Speeds

Reader p. 13-14

Compare the _____ ball at two speeds.

A bigger _____ is how we know the faster ball had more energy of motion.

BALL → **OBSERVE** → **EVIDENCE**

Preview: one ball, two setups. Which cup move was bigger?

N2 What We Measured

Teacher Guide Lesson 4 · AP 4.1

We measure ramp _____ in centimeters.

We measure how far the _____ moves in centimeters.

We do not measure a speed number.

The one thing we change is ramp _____.

We keep the _____ ball.

Record the nine measured trials on the Ramp It Up sheet. The first roll is a calibration trial. Do not record it.

Which setup do you predict will move the cup farther? Why?

_____

_____

What change did you measure?

_____

_____

N3 Steeper Ramp

Reader p. 15

Objects pick up more _____ on steeper ramps.

From the steeper ramp, that ball made a _____ change.

N4 The Bounce

Reader p. 16

The faster ball bounced _____ than the slower ball.

A louder sound is also a bigger change you can observe.

CHECK Your Thinking

How are energy and speed related? Name the ball you kept and one change that was bigger.

BQ · NOW I THINK How are energy and speed related?